

# claude-windows / 45ea3a91-654d-4972-9cd3-65f35db54caf

## Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\45ea3a91-654d-4972-9cd3-65f35db54caf\subagents\workflows\wf_2a64bc88-2a8\agent-a3c48bdd13621de96.jsonl`  
- SHA-256: `161444160199f52e17f0c4e03fa8bd2ab6b9a4209a5c1c1ec3b95b79fd72b228`  
- Source modified: `2026-06-10T05:44:02+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_2a64bc88-2a8`  
- session_id: `45ea3a91-654d-4972-9cd3-65f35db54caf`
```

## Transcript

### 1. user (2026-06-10T05:40:49.621Z)

CONTEXT ? two sibling repos on a Windows machine:

- INDEXER: C:/Users/avidu/Projects/badminton-highlight-indexer ? local AI badminton(->racket-sports) highlight indexer + rally detector. FastAPI + SQLite + ffmpeg + GPU CV via WSL; Gemini = paid cloud AI. docs/ tree is rich. Current branch docs/next-steps-multisport (master = main branch).

- COLLECTOR: C:/Users/avidu/Projects/sports-data-collector ? golden-label acquisition/curation/ingestion CLI (system-of-record candidate for training corpus).

Project state (June 2026): scaffolding complete; owned-model roadmap agreed (docs/OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md) but NO platform/owned-model implementation started; next committed platform slice = async job model. Licensing guardrail: MIT/Apache-2.0/BSD only for code+data+weights; paid LLMs are inference-only, never training-data sources. ENVIRONMENT RULES: this dev environment has NO GPU/torch/cv2 ? do NOT attempt to run the video pipeline or import torch/cv2 modules. Test suites are offline/fully-mocked and SAFE to run. You are READ-ONLY: never edit/create/delete repo files; running tests and git read commands is allowed.

QUALITY BAR: cite file paths (and line numbers where possible) for every finding. Report what IS in the code/docs, not what you assume. Be skeptical and concrete. Your final structured output is consumed programmatically by an orchestrator ? return raw data, not prose for a human.

ROLE: skeptical reviewer of a PMF research report. Below is the report. Check its 4-5 most load-bearing market claims with quick web searches (load WebSearch via ToolSearch if needed); flag any unsupported, stale, or cherry-picked claims; flag motivated reasoning in the verdict; name anything important it MISSED. Return a concise plain-text critique (bullet list).

REPORT: {"summary":"Internal thesis (docs/PMF\_MARKET\_RESEARCH.md, docs/COMMERCIALIZATION.md §6, docs/PLATFORM\_ARCHITECTURE.md): beachhead = India (Bangalore) badminton coaching academies; wedge = software-only BYO-camera upload of full static-camera match video -> auto rally segmentation + highlight reels, evolving to conversational video-QA 'assistant coach' and multisport racquet; pricing anchors B2C \$60-180/yr, B2B academy \$300-1,500/yr (PMF\_MARKET\_RESEARCH.md:46-48); cost model Gemini 2.5 Flash ~\$0.05 input per 10-min clip (COMMERCIALIZATION.md:122-124); claimed whitespace 'incumbents ignore badminton, only direct competitor BadPro+ is early' (PMF\_MARKET\_RESEARCH.md:160-164). External research VERIFIES the pricing/cost benchmarks (SwingVision \$179.99/yr, PB Vision \$99/yr, Trace \$180-300/yr, Gemini Flash \$0.30/1M input live) but FALSIFIES the India-whitespace claim: at least four funded, celebrity-distributed Indian/Bangalore badminton AI players exist (Machaxi - \$1.5M from Rainmatter + Prakash Padukone; VISIST.AI - 70+ Bengaluru academies, Padukone School partnership; Game Theory - Gopichand-backed smart courts with auto highlights; Stupa Sports Analytics - \$3.3M, BWF-approved badminton IRS), none of which appear in the internal competitive table. The residual defensible wedge is narrower than the docs claim: indexing EXISTING messy multi-court academy footage (a problem the market only solves with per-court installed hardware), not 'badminton AI for India' generically.", "market\_landscape":[{"claim":"SwingVision Pro is \$14.99/mo or \$179.99/yr, tennis + pickleball only; no badminton support today. Confirms internal pricing benchmark exactly (PMF\_MARKET\_RESEARCH.md:139).", "source":"https://swing.vision/subscribe", "relevance":"Closest analog; validates the \$180/yr consumer ceiling for racket-sport AI

software."},{ "claim": "SwingVision has ~20,000 paying subscribers, claims \$4M+ ARR; FY2024 revenue \$2.75M, up only ~12% YoY from \$2.46M (2023); raised \$10M+ total." , "source": "https://kingscrowd.com/swingvision-on-wefunder-2025/" , "relevance": "WTP exists but the closest analog grows slowly ? tempers B2C upside assumptions; also implies paid conversion ~10% of its claimed 200k monthly users." } , { "claim": "SwingVision raised \$6M explicitly to expand into pickleball and padel; expansion to other racket sports incl. badminton described as applicable/on roadmap." , "source": "https://www.sportspro.com/news/swingvision-ai-tennis-app-pickleball-padel/" , "relevance": "Internal risk #5 (competitive response, PMF\_MARKET\_RESEARCH.md:201-203) is live, not hypothetical." } , { "claim": "BadPro+ (founded by an Olympic badminton athlete + Microsoft engineer) sells pay-per-match AI badminton analysis with no subscription, plus federation and per-call API tiers." , "source": "https://www.badproplusapp.com/" , "relevance": "Confirms internal read (PMF\_MARKET\_RESEARCH.md:140): real badminton AI demand signal but unproven subscription WTP." } , { "claim": "Machaxi (Bengaluru) raised \$1.5M in 2025 led by Rainmatter (Zerodha) with Prakash Padukone participating; AI-powered badminton coaching centers; plans 1,000+ centres with Padukone School of Badminton across India." , "source": "https://india.entrepreneur.com/news-and-trends/machaxi-secures-15-million-to-expand-ai-powered-badminton/493070" , "relevance": "Funded, celebrity-distributed badminton AI in the exact beachhead city ? absent from internal competitive table." } , { "claim": "VISIST.AI (Bengaluru, IISc-incubated) does AI video analysis of badminton technique/skill assessment; 70+ Bengaluru academies use it; partnered with Padukone School of Badminton." , "source": "https://theprint.in/india/visist-ai-partners-with-padukone-school-of-badminton-to-digitise-athlete-skill-assessments/2716417/" , "relevance": "Directly contradicts 'incumbents ignore badminton' for the #1 beachhead; owns academy relationships the product would need. Note: skill assessment, not full-match rally segmentation/highlights." } , { "claim": "Game Theory (Bengaluru, Gopichand/Bopanna/Sharath Kamal investors, Rainmatter-backed) runs smart courts across badminton/pickleball/padel/TT with built-in recording and 'Instagram-ready video highlights' delivered in-app after booked games." , "source": "https://yourstory.com/2024/02/computer-vision-game-theory-sports-tech-startup-nithin-kamath-rainmatter" , "relevance": "Amateur highlight delivery already happening in Bangalore ? bundled with court booking, implying highlights alone may be a feature, not a product." } , { "claim": "Stupa Sports Analytics (India) raised INR 28 Cr (~\$3.3M) pre-Series A; started in table tennis, expanded to badminton; its AI instant-review system is BWF-approved; serves 15+ federations." , "source": "https://inc42.com/buzz/stupa-sports-analytics-nets-inr-28-cr-to-offer-ai-solutions-to-sports-federations/" , "relevance": "Indian racket-sports CV company with federation distribution; also validates the repo's table-tennis transfer finding (NEXT\_STEPS.md:20-23) as a real market." } , { "claim": "PB Vision (pickleball AI) charges \$99/yr (~100 min analysis/mo) or \$19.99/mo; its partner API is priced at \$8 per hour of HD video processed." , "source": "https://pb.vision/" , "relevance": "Sits at the BOTTOM of the internal B2C band (\$60-180/yr); the \$8/hr API price is a direct external benchmark for the planned hosted per-video API." } , { "claim": "Veo Cam 3 ~\$1,533 hardware + Team \$49/mo / Club \$149/mo subscriptions; Trace is \$180/yr (Basic) - \$300/yr (Pro) with free leased camera." , "source": "https://www.veo.com/pricing and https://traceup.com/pricing" , "relevance": "Verifies internal hardware-bundled benchmarks (PMF\_MARKET\_RESEARCH.md:134-136) are current." } , { "claim": "Baseline Vision sells a \$2,000-2,300 portable AI tennis camera (line calling + analytics)." , "source": "https://www.baselinevision.com/product" , "relevance": "Hardware price point amateurs do pay in tennis; reinforces that capture hardware, not software, is where incumbents monetize." } , { "claim": "Padel already has multiple AI-camera highlight vendors: Wingfield (tennis/padel/pickleball, used by Rafael Nadal Academy), Clutch ('AI camera for padel - automated highlights'), Padelytics, PlaySight padel SmartCourt." , "source": "https://www.wingfield.io/en and https://www.clutchapp.io/" , "relevance": "Racket-sport auto-highlights is a crowded category in padel; differentiation must be sport (badminton) + footage type (BYO multi-court), not the highlight concept." } , { "claim": "WSC Sports (550+ orgs: NBA, NHL, LALIGA, DAZN) and Magnifi/Reely sell enterprise broadcast highlight automation with opaque custom pricing; Reely targets esports; none offer amateur self-serve for court sports." , "source": "https://wsc-sports.com/ and https://magnifi.ai/products/reely" , "relevance": "Confirms internal claim that broadcast highlight giants do not serve the amateur static-camera segment." } , { "claim": "The 'multi-court' problem is solved in-market only by per-court installed hardware (PlaySight SmartCourt ? up to 24 cameras; Hudl Focus; Pixellot); no software-only product found that handles adjacent-court contamination in a single static camera view." , "source": "https://playsight.com/ and https://www.hudl.com/products/focus" , "relevance": "The repo's multi-court robustness finding (NEXT\_STEPS.md:23-26) targets a genuinely unsolved software gap ? but the market's revealed alternative is hardware-per-court." } , { "claim": "Gemini 2.5 Pro 'rivals specialized fine-tuned models' on video benchmarks incl. QVHighlights moment retrieval and YouCook2 dense captioning,

```
frames.", "source": "https://developers.googleblog.com/en/gemini-2-5-video-
```

technically-grounded differentiator in the whole thesis.", "MISSING ENTIRELY from internal docs: (a) who physically captures the video at an academy (existing CCTV vs phone-on-tripod vs new hardware) and who owns that workflow; (b) who pays ? player vs parent vs coach vs academy owner ? Indian comparables monetize via coaching/venue fees, not analysis SaaS; (c) celebrity/federation distribution as the apparent dominant GTM in Indian badminton; (d) any demand evidence for the conversational video-QA differentiator (PLATFORM\_ARCHITECTURE.md §5A invests heavily in it with zero customer signal); (e) the C13 beachhead-validation interviews (COMMERCIALIZATION.md:40) remain not done ? the entire beachhead is desk research."], "opportunities": ["The confirmed gap: software-only, BYO-footage, full-match rally indexing + highlights for badminton ? especially on messy MULTI-COURT academy footage, which no vendor handles in software and incumbents solve with per-court hardware costing \$949-1,533/camera + subscriptions per court. For a 6-12 court Indian academy, a single existing camera + software is an order of magnitude cheaper.", "Partner-not-compete distribution in India: VISIST.AI (technique assessment) and Machaxi (coaching centers) lack rally-segmentation/highlight tech; a white-label or API integration (PB Vision x PodPlay is the proven template ? <https://podplay.app/blog/podplay-x-pb-vision-raising-the-ai-video-analysis-bar-for-pickleball-clubs>) converts the celebrity-distribution threat into a channel.", "API pricing surface validated externally: PB Vision charges partners \$8/hr of HD video processed; with Gemini COGS ~\$0.30-0.60/hr (or lower self-hosted), a per-hour processing API matches the repo's hosted-API architecture (PLATFORM\_ARCHITECTURE.md) and avoids the consumer-app CAC war.", "Table tennis is an underrated second sport: the repo's own WASB-transfer finding (TT F1 0.909 out-of-the-box, NEXT\_STEPS.md:20-23) aligns with Stupa proving an Indian TT analytics market with federation buyers; TT clubs/federations may be a less contested beachhead than badminton academies.", "BadPro+'s pay-per-match model (no subscription) is a tested monetization shape for price-sensitive India; per-video pricing fits the upload-API product better than SaaS tiers and lowers the WTP validation bar.", "Padel/pickleball vendors (Wingfield, Clutch) prove auto-highlight demand in racket sports but are all hardware-attached; a camera-agnostic engine could license INTO those ecosystems."], "risks": ["Distribution lock-up: Padukone (VISIST.AI, Machaxi) and Gopichand (Game Theory) networks are absorbing the credible paths into Indian badminton academies now, while this project has shipped no customer-facing product and done zero discovery interviews (C13, COMMERCIALIZATION.md:40 still 'Unvalidated').", "Commoditization clock: coarse highlights from long video are already a thin Gemini wrapper (\$0.30-0.50/1M video input tokens); if Gemini-class models close the precise-boundary gap before the owned model ships, the moat reduces to dataset + distribution ? and the project has little of the latter.", "SwingVision (\$10M+ raised, 20k paying subs) raised \$6M specifically to go multi-racket-sport; padel is explicit, badminton mentioned as applicable ? a direct collision in the prosumer segment within the planning horizon.", "Capture-behavior assumption untested: the entire B2C funnel assumes amateurs put a phone on a tripod; every funded competitor concluded the opposite and installed hardware or owns venues. Internal risk #1 (PMF\_MARKET\_RESEARCH.md:189-191) remains the thesis-killer and is cheap to test but untested.", "WTP compression in India: Game Theory delivers highlights bundled with court booking; group coaching is INR 1,500-5,000/mo; a \$600/yr (~INR 50k) academy SaaS must displace tangible coaching budget without a celebrity endorsement.", "Closest-analog growth is slow: SwingVision +12% YoY revenue at \$2.75M after ~8 years suggests this category produces niche businesses, not venture-scale ones ? fine for a solo/lifestyle business, fatal for a 'platform' ambition if that's the goal.", "Solo, pre-product vs funded field: Machaxi \$1.5M, Stupa \$3.3M, SwingVision \$10M+, Game Theory (Rainmatter) ? all shipping today; the repo is at 'scaffolding complete, no platform implementation started' (CLAUDE.md current state).", "Internal research staleness process risk: the 2026-06-07 market study missed every Indian competitor despite India being its own #1 recommendation ? competitive scanning needs to be a recurring gate, not a one-shot snapshot."], "pmf\_verdict": "The thesis half-survives. What the evidence supports: real WTP exists for racket-sport AI video at \$99-180/yr consumer and per-court/per-hour B2B (SwingVision 20k paying subs; PB Vision \$99/yr + \$8/hr API; Trace/Veo club tiers); the cost model is verified and leaves wide margin; and the single most defensible claim in the repo is the one its own experiments produced ? nobody in the market handles multi-court, adjacent-game-contaminated static footage in software, while the market's only answer (per-court installed cameras) is expensive at academy scale. What the evidence breaks: the 'India badminton whitespace' framing is wrong ? Bangalore specifically is the most contested badminton-AI city on earth right now (VISIST.AI in 70+ local academies via Padukone, Machaxi funded by Padukone/Rainmatter, Game Theory by Gopichand, Stupa with BWF approval), and all of them own the distribution this project lacks. The strongest honest wedge is therefore not 'badminton AI for Indian academies' but 'rally-precise indexing of existing messy academy footage, sold per-hour as an engine/API' ? possibly THROUGH the very players that look like competitors, none of whom have this capability.\n\nFastest falsification path, in order, before any more engineering: (1) run the already-planned-but-undone C13 discovery interviews (10-15 Bengaluru academies) with two added questions ? do you already have multi-

court footage nobody watches, and would you pay per-match to have it indexed ? this tests both the wedge and the capture assumption in two weeks for \$0; (2) run a Gemini-3-Flash-wrapper baseline against the human-GT LOVO corpus (the harness already exists in backend/eval/) ? if a \$0.50/1M wrapper hits within a few F1 points of the learned prototype's 0.583, the owned-model investment is premature and the play is distribution/UX, not modeling; (3) one paid pilot with a single academy using their existing footage at a real price (even INR 500/match) beats any further desk research. The verdict: a viable niche product with a genuine, narrow technical moat and a verified cost structure ? but currently mis-aimed at a beachhead whose distribution is being captured by funded incumbents, with every demand-side assumption still resting on zero customer contact. Validate demand before the async-job slice or any owned-model milestone consumes another engineering month."}

## 2. user (2026-06-10T05:41:06.830Z)

No files found

## 3. user (2026-06-10T05:41:13.348Z)

BACKLOG.md  
CODE\_MAP.md  
COMMERCIALIZATION.md  
DEFERRED\_HARDENING\_PLAN.md  
EVALUATION.md  
GOLDEN\_SET\_IMPLEMENTATION.md  
GOLDEN\_SET\_PHASE1\_VIDEOS.md  
GOLDEN\_SET\_VENDORING.md  
HOW\_RALLY\_DETECTION\_WORKS.md  
NEXT\_STEPS.md  
OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md  
PLATFORM\_ARCHITECTURE.md  
PMF\_MARKET\_RESEARCH.md  
README.md  
ROADMAP.md  
TRACKNET\_WSL\_SETUP.md  
VIDEO\_RECORDING\_GUIDELINES.md  
archives

## 4. assistant (2026-06-10T05:41:19.617Z)

Now let me verify the internal citations in the report against the actual docs.

## 5. user (2026-06-10T05:41:20.606Z)

```
1      # PMF & Market Study ? AI Highlight-Reel Tool for Self-Recorded Racket Sports
2
3      > **Research snapshot ? 2026-06-07.** Figures are cited as-of with source + year;
4      > validate live before relying on them for investment/pricing decisions. This is the
5      > market-research-of-record behind the PMF section of the living
6      > [`COMMERCIALIZATION.md`](COMMERCIALIZATION.md) tracker ($6). Product framing:
7      > a closed-source hosted API/SaaS (plus local mode) that ingests long-form,
8      > self-recorded, **static-camera, full-court amateur** racket-sports video and
9      > auto-detects rallies and compiles highlight reels. Primary sport: badminton;
10     > sport-generic architecture roadmapped to tennis, pickleball, table tennis, cricket.
11
12     > **Confidence convention.** Figures from named industry reports, federations, or
13     > company pricing pages are tagged with source + year. Inferred items are tagged
14     > **(estimate)** / **(low confidence)**. Emerging-category market sizes vary widely
15     > between vendors; ranges are shown rather than false precision.
16
17     ---
18
19     ## 1. Executive summary
20
21     The product sits at the intersection of three favorable trends: (a) a fast-growing
```

22 **\*\*sports analytics market\*\*** (~USD 5.76B in 2025, ~18% CAGR to USD 24.36B by ~2034  
23 depending on vendor); (b) a structural shift to **\*\*amateur self-recording\*\*** on action  
24 cameras and phones (action-camera market ~USD 6.5B in 2024, sports the largest  
25 segment); and (c) **\*\*strong racket-sport participation\*\*** ? 220M+ badminton players,  
26 106M tennis players (2024), and a near-vertical pickleball curve (~19.8M US players in  
27 2024, +45.8% YoY).

28  
29 The central tension is the **\*\*static-camera constraint\*\***: the tool works best on fixed,  
30 full-court amateur footage and **\*fails\*** on broadcast/panning footage. This is both a  
31 moat (most competitors target team sports / broadcast) and a filter on the addressable  
32 user ? the customer must already set a camera on a tripod behind the court. That  
33 behavior is concentrated in **\*\*coaches/academies, clubs, and serious enthusiasts\*\***, not  
34 the median casual player.

35  
36 **\*\*Recommended beachhead (strongest single):\*\*** **\*\*badminton coaching academies and**  
37 **serious club/competitive players in India\*\*** ? a large, rapidly formalizing academy  
38 ecosystem (Khelo India tailwind), genuine fixed-angle recording behavior, existing  
39 willingness to pay for coaching, an underserved tooling gap (incumbents are  
40 team-sports/broadcast), and a direct-but-immature competitor field. The product is  
41 **\*native badminton-first\*** ? a genuine edge here. Secondary: **\*\* (2) US/global pickleball**  
42 **players & clubs\*\*** (capital-rich, fastest-growing, but SwingVision already strong);  
43 **\*\* (3) badminton academies/clubs in East/SE Asia + Denmark\*\*** (highest density).

44  
45 **\*\*Pricing\*\*** clusters into hardware-bundled auto-filming (Veo, Trace, Pixellot, Spiideo)  
46 and software-only phone AI (SwingVision USD 179.99/yr; BadPro+ pay-per-use). A  
47 software-only BYO-camera model undercuts the hardware players; realistic anchors  
48 **\*\* (estimate): B2C USD 60-180/yr; B2B academy/club USD 300-1,500/yr per location.\*\***

49  
50 **\*\*Top risks:\*\*** (1) amateurs won't reliably set up a tripod ? TAM collapses to  
51 coaches/clubs; (2) free/near-free alternatives cap B2C WTP; (3) capture-quality  
52 variance is the core failure mode; (4) badminton's strongest markets (India, Indonesia,  
53 China) are the most price-sensitive and hardest to bill at Western SaaS prices.

54  
55 ---  
56

## 57 ## 2. Market sizing & trends

### 58 59 ### 2.1 Sports analytics / sports-video market

- 60 - **\*\*2025 size:\*\*** ~USD 5.47B (Precedence Research) to ~USD 5.79B (Fortune Business  
61 Insights); other vendors USD 5.676.0B. **\*(2025)\***
- 62 - **\*\*Growth:\*\*** CAGR 15.7-22.5%. Precedence ? ~USD 29.75B by 2034 (20.63%); Fortune BI  
63 ? USD 24.03B by 2032 (22.5%); MarketsandMarkets (narrower) USD 4.75B by 2030 (15.7%);  
64 Future Market Insights ? ~USD 36.2B by 2035 (22.1%). **\*(reports 2024-2025)\***
- 65 - **\*\*Video analytics is the leading segment\*\*** within sports analytics (Grand View /  
66 Fortune BI) ? the directly relevant sub-market.
- 67 - **\*\*Interpretation (estimate):\*\*** the **\*amateur/club auto-highlight\*** slice is a small but  
68 fast-growing fraction; most analytics revenue today is pro/broadcast/team-sports. The  
69 product targets the long tail the big numbers under-serve.

### 70 71 ### 2.2 Action-camera / amateur recording adoption

- 72 - **\*\*2024 size:\*\*** ~USD 6.5B (GMIInsights) ? ~USD 16.9B by 2034 (~10% CAGR). **\*(2024-2025)\***
- 73 - **\*\*Sports is the largest end-use segment\*\*** of action cameras (GMIInsights, 2024).
- 74 - **\*\*GoPro ~47% share, DJI ~20%\*\*** (electroIQ/industry, 2024); phones + cheap tripods are  
75 the de-facto amateur rig.
- 76 - **\*\*Implication:\*\*** the capture-side behavior the product depends on is already  
77 widespread and growing ? **\*for the segments that record\*** ? validating "BYO-camera,  
78 software-only."

### 79 80 ### 2.3 Participation & popularity by sport and geography

81  
82 | Sport | Participation signal | Source (year) |  
83 |---|---|---|  
84 | **\*\*Badminton\*\*** | 220M+ play regularly; 160+ countries | BWF est. via BadmintonBites  
(2023) |  
85 | Badminton ? China | ~80M registered, up to ~100M (largest base) | industry

compilations (2023) |

86 | Badminton ? India | 20M+ recreational by 2023; academy boom + Khelo India | industry compilations (2023) |

87 | Badminton ? Indonesia | National sport; ~1M club players | industry (est.) |

88 | Badminton ? Denmark | 100,000+ registered club members (Europe #1) | industry (2023) |

89 | Badminton ? Japan/Korea | Major school-club + workplace sport | industry (2023) |

90 | **Tennis** | **106M** players (2024), +25.6% over 5 yrs (87M in 2021) | ITF Global Tennis Report (2021, 2024) |

91 | Tennis ? Asia | ~1/3 of players (33M of 87M) | ITF (2021) |

92 | **Pickleball** ? US | **19.8M** players (2024), +45.8% YoY, +311% over 3 yrs; ~6.2M core | SFIA / Pickleheads / The Dink (2024) |

93 | Pickleball ? US courts | ~70,000 courts, +55% places to play YoY | SFIA / Pickleheads (2024) |

94 | Pickleball ? global | IFP 78 member countries; UK +73% in 2024; ~282M played monthly per PPA claim **(low confidence)** | various (2024?2025) |

95 | **Table tennis** | 30M+ players; 227 member associations | ITTF (current) |

96 | **Cricket** | 2.5B+ fans; 500M+ in India (mostly **\*fans\***, not self-recorders) | Cricket Rise / Nielsen (2024?2025) |

97

98 - **Pickleball market value:** ~USD 2.2B (2024) ? USD 9.1B by 2034 at 15.3% CAGR

99 (Market.us/Custom Market Insights, 2024?2025).

100 - **Badminton equipment market:** ~USD 7.23B global (2023), Asia-Pacific dominant.

101

102 **Geographic read:** badminton density is overwhelmingly Asian + Denmark; tennis is broad/high-income; pickleball is US-dominant, capital-rich, globalizing fast; cricket

103 is huge as **\*fandom\*** but its amateur self-recording + static-full-court fit is weak

104 (large field, not a fixed-frame court). For **\*this product's static-camera fit\***,

105 badminton/pickleball/tennis/table-tennis are the core; cricket is a long-tail roadmap

106 item, not a beachhead.

107

108

109 ---

110

111 **## 3. Customer segment evaluation**

112

113 | Segment | Market-size signal | Willingness to pay | Accessibility | Static-camera fit | B2C/B2B | Verdict |

114 | ---|---|---|---|---|---|---|

115 | **Individual amateur/enthusiast** | Largest raw TAM (100s of millions) | **Low** ? competes with free; ~USD 5?15/mo ceiling | High via app stores; high CAC/churn | **Mixed** ? most won't set a tripod | B2C | Volume play, weak economics; use as funnel |

116 | **Coaches & coaching academies** | India academy boom; thousands across Asia | **High** ? already sell coaching; USD 300?1,500/yr plausible | Reachable via federations/distributors/social | **Strong** ? already film fixed-angle | B2B/B2B2C | **Best fit** ? buyer, recorder, payer aligned |

117 | **Clubs & leagues** | Denmark 100k+; thousands of clubs Asia/EU | Medium-high; USD 500?3,000/yr | Medium ? committee sales cycle | **Strong** | B2B/B2B2C | Strong secondary; longer cycle |

118 | **Tournament/event organizers** | Many small/regional events | Medium ? per-event/seasonal | Medium ? episodic | **Strong** | B2B | Good per-event upsell; lumpy |

119 | **Schools & universities** | Large (esp. Asia/Japan clubs) | Low-medium ? budget-constrained | Low ? slow procurement | Strong | B2B | Slow; grant/Khelo-India dependent |

120 | **National/regional federations** | Few, high-value | Medium-high; procurement-heavy/political | Low ? long cycles | Strong | B2B | Lighthouse/credibility, not beachhead |

121 | **Broadcasters / streamers** | High value per deal | High | Low ? incumbents entrenched | **Poor** ? panning = failure mode | B2B | **Avoid** ? violates core constraint |

122

123 **Key dynamics:** the static-camera constraint is a **B2B filter** ? coaches/academies/

124 clubs already record from a fixed angle, so the product fits existing behavior (cleanest

125 PMF). B2C is a large low-WTP, high-CAC funnel best used to feed B2B. Broadcasters

126 explicitly violate the constraint ? exclude.

127

128 ---

129

130 **## 4. Competitive & pricing benchmarks**

131

|     | Product                                                                                 | Category                                | Model                   | Price                                        | Notes (year)                                                                  |
|-----|-----------------------------------------------------------------------------------------|-----------------------------------------|-------------------------|----------------------------------------------|-------------------------------------------------------------------------------|
| 132 |                                                                                         |                                         |                         |                                              |                                                                               |
| 133 |                                                                                         |                                         |                         |                                              |                                                                               |
| 134 | **Veo (Cam 3)**                                                                         | Team-sports auto-filming                | Hardware + sub          | Camera ~USD 1,533 (5G ~1,886) + ~£395?895/yr | 15,000+ clubs/80 countries; \$115M raised (2025)                              |
| 135 | **Trace**                                                                               | Soccer auto-filming                     | **Free camera + sub**   | USD 180/yr (Basic) ? 300 (Pro)               | Camera leased free with sub (2025)                                            |
| 136 | **Pixellot (Air)**                                                                      | Auto camera/streaming                   | Hardware + sub          | Camera ~USD 949 + USD 69?167/mo              | Institutional/league; 10,000+ clubs (2025)                                    |
| 137 | **Spiideo**                                                                             | Auto-filming/analytics                  | Sub incl. camera        | ~USD 3,800/yr (1 team) ? 6,100/yr (club)     | Institutional budgets (2025)                                                  |
| 138 | **Hudl**                                                                                | Team video analysis                     | SaaS tiers              | From ~USD 400/yr                             | Broad team-sports incumbent (2026)                                            |
| 139 | **SwingVision**                                                                         | **Tennis/pickleball AI, single iPhone** | Software freemium       | **USD 179.99/yr**                            | free 2 hrs/mo   Closest analog; 200k+ monthly users; \$10M+ raised (2025)     |
| 140 | **BadPro+**                                                                             | **Badminton AI on phone**               | **Pay-per-use, no sub** | Per-analysis (? "< one coaching session")    | Direct badminton competitor; shot detection, heatmaps, auto-highlights (2025) |
| 141 |                                                                                         | Niche pickleball/badminton apps         | Various                 | Mixed                                        | Varies   Fragmented; few true auto-highlight tools (2025)                     |
| 142 |                                                                                         |                                         |                         |                                              |                                                                               |
| 143 | **What buyers actually pay (synthesis):** consumer software-only anchors at ~USD        |                                         |                         |                                              |                                                                               |
| 144 | 100?200/yr (SwingVision 179.99); hardware-bundled USD 180?900/yr prosumer/club,         |                                         |                         |                                              |                                                                               |
| 145 | USD 3,800?6,100/yr institutional; badminton-specific BadPro+ uses *pay-per-use* ?       |                                         |                         |                                              |                                                                               |
| 146 | signaling a real badminton AI market but **unproven subscription WTP** (opportunity +   |                                         |                         |                                              |                                                                               |
| 147 | warning). **Positioning:** software-only + BYO-camera undercuts hardware players and    |                                         |                         |                                              |                                                                               |
| 148 | sells worldwide via SaaS. Realistic anchors **(estimate): B2C USD 60?180/yr; B2B        |                                         |                         |                                              |                                                                               |
| 149 | academy/club USD 300?1,500/yr** per location. The differentiators (cheap offline        |                                         |                         |                                              |                                                                               |
| 150 | self-improving detection, per-rally stats/tagging, clean licensing) support a **B2B**   |                                         |                         |                                              |                                                                               |
| 151 | value story more than a B2C one.                                                        |                                         |                         |                                              |                                                                               |
| 152 |                                                                                         |                                         |                         |                                              |                                                                               |
| 153 | ---                                                                                     |                                         |                         |                                              |                                                                               |
| 154 |                                                                                         |                                         |                         |                                              |                                                                               |
| 155 | ## 5. Recommended beachhead                                                             |                                         |                         |                                              |                                                                               |
| 156 |                                                                                         |                                         |                         |                                              |                                                                               |
| 157 | Ranked by **WTP × participation density × existing recording behavior × static-camera   |                                         |                         |                                              |                                                                               |
| 158 | fit × competitive whitespace.**                                                         |                                         |                         |                                              |                                                                               |
| 159 |                                                                                         |                                         |                         |                                              |                                                                               |
| 160 | ### #1 (strongest) ? Badminton coaching academies & serious players, **India**          |                                         |                         |                                              |                                                                               |
| 161 | 20M+ recreational players (2023); rapidly **formalizing academy ecosystem** + Khelo     |                                         |                         |                                              |                                                                               |
| 162 | India tailwind; academies already record fixed-angle; coaches are buyers who monetize   |                                         |                         |                                              |                                                                               |
| 163 | instruction (high relative WTP); incumbents ignore badminton; the only direct badminton |                                         |                         |                                              |                                                                               |
| 164 | AI (BadPro+) is early with unproven pay-per-use. Product is *native badminton-first*.   |                                         |                         |                                              |                                                                               |
| 165 | **Watch-outs:** price sensitivity (need INR-tuned tiers); sell to the **academy**, not  |                                         |                         |                                              |                                                                               |
| 166 | the individual.                                                                         |                                         |                         |                                              |                                                                               |
| 167 |                                                                                         |                                         |                         |                                              |                                                                               |
| 168 | ### #2 ? Serious pickleball players & clubs, **US (then UK/Australia/Asia)**            |                                         |                         |                                              |                                                                               |
| 169 | Fastest-growing sport (19.8M US, +45.8% YoY, 2024); capital-rich; ~70,000 courts;       |                                         |                         |                                              |                                                                               |
| 170 | strong recording/sharing culture; pickleball is on the roadmap. **Watch-out:**          |                                         |                         |                                              |                                                                               |
| 171 | **SwingVision already owns single-camera tennis+pickleball AI** at USD 179.99/yr (200k+ |                                         |                         |                                              |                                                                               |
| 172 | monthly users) ? head-to-head differentiation, not greenfield.                          |                                         |                         |                                              |                                                                               |
| 173 |                                                                                         |                                         |                         |                                              |                                                                               |
| 174 | ### #3 ? Badminton academies/clubs, **East/SE Asia (China, Indonesia, Malaysia, Japan,  |                                         |                         |                                              |                                                                               |
| 175 | Korea) + Denmark**                                                                      |                                         |                         |                                              |                                                                               |
| 176 | Highest global density (China 80M+; Denmark 100k+ structured club members; Japan/Korea  |                                         |                         |                                              |                                                                               |
| 177 | school-club & workplace systems), mature club infra, fixed-court recording.             |                                         |                         |                                              |                                                                               |
| 178 | **Watch-outs:** China needs local hosting/partners + is regulatory-complex; SE Asia     |                                         |                         |                                              |                                                                               |
| 179 | price-sensitive; Denmark/EU small but high-WTP and easy to pilot.                       |                                         |                         |                                              |                                                                               |
| 180 | **Bottom line:** lead with **India badminton academies** (whitespace + behavior + buyer |                                         |                         |                                              |                                                                               |
| 181 | alignment), use **enthusiast B2C as the funnel**, validate a **club/academy B2B2C       |                                         |                         |                                              |                                                                               |
| 182 | motion**, then expand to **pickleball-US** (out-build SwingVision) and                  |                                         |                         |                                              |                                                                               |
| 183 | **badminton-Asia/Denmark** (density).                                                   |                                         |                         |                                              |                                                                               |
| 184 |                                                                                         |                                         |                         |                                              |                                                                               |
| 185 | ---                                                                                     |                                         |                         |                                              |                                                                               |
| 186 |                                                                                         |                                         |                         |                                              |                                                                               |



```

187  ## 6. Risks & open questions (what would invalidate the thesis)
188
189  1. **Tripod/static-camera adoption among amateurs.** If serious amateurs won't set up a
190     fixed camera, B2C TAM collapses to coaches/clubs. *Validate:* observe 3?5 academies +
191     50?100 enthusiasts on whether they already record fixed-angle.
192  2. **Free / near-free alternatives** (YouTube, free app tiers, BadPro+ pay-per-use) cap
193     B2C WTP. *Validate:* fake-door pricing at USD 60?180/yr.
194  3. **Capture-quality variance = the failure mode** (gym lighting, off-center angle,
195     occlusion, mixed resolutions). *Validate:* benchmark rally-detection P/R on 100+ real
196     amateur clips; define a "good-capture" spec + auto-reject (ties to
197     `VIDEO_RECORDING_GUIDELINES.md` + the observability reports).
198  4. **Monetization mismatch in best-participation markets** (India/Indonesia/China are
199     price-sensitive, hard to bill). *Validate:* local pricing/payments pilot; per-academy
200     INR licensing.
201  5. **Competitive response** (SwingVision or a team-sports incumbent adds racket sports).
202     *Validate:* ship a badminton-quality advantage incumbents can't trivially match
203     (offline self-improving detection, per-rally tagging).
204  6. **B2B sales-cycle drag** for clubs/federations/schools. *Validate:* single-coach/
205     academy self-serve before institutional deals.
206  7. **Clean-licensing positioning** matters mainly to B2B buyers ? confirm it's a real
207     purchasing criterion via 5?10 buyer interviews.
208
209  **Cheapest next steps:** (a) 10?15 discovery interviews with Indian badminton academy
210  owners/coaches; (b) a capture-quality benchmark on real amateur footage; (c) a fake-door
211  pricing test across B2C (~USD 99/yr) and B2B academy (~USD 600/yr).
212
213  ---
214
215  ## 7. Sources
216
217  - Sports analytics market: Precedence Research; Fortune Business Insights; Grand View
218    Research; MarketsandMarkets; Future Market Insights. *(2024?2025)*
219  - https://www.precedenceresearch.com/sports-analytics-market
220  - https://www.fortunebusinessinsights.com/sports-analytics-market-102217
221  - https://www.grandviewresearch.com/industry-analysis/sports-analytics-market
222  - https://www.marketsandmarkets.com/PressReleases/sports-analytics.asp
223  - https://www.futuremarketinsights.com/reports/sports-analytics-market
224  - Action camera market: https://www.gminsights.com/industry-analysis/action-camera-
225    market (2024);
226    https://www.precedenceresearch.com/action-camera-market (2024);
227    https://electroi.com/stats/action-camera-statistics/ (2024)
228  - Badminton participation/market: https://badmintonbites.com/how-many-badminton-players-
229    are-there-in-the-world/ ;
230    https://stringsandpaddles.com/where-is-badminton-played-10-countries-where-its-
231    popular/ ;
232    https://en.wikipedia.org/wiki/Badminton ;
233    https://www.marketgrowthreports.com/market-reports/badminton-market-110049 ;
234    https://www.thesportsschool.com/top-10-badminton-academies-in-india/
235  - Tennis: https://www.itftennis.com/en/news-and-media/articles/itf-global-tennis-report-
236    participation-hits-106-million-in-five-years/ ;
237    https://sgbonline.com/itf-global-tennis-report-2021-shows-worldwide-participation-
238    gains/
239  - Pickleball: https://sfia.org/resources/as-pickleball-continues-unprecedented-growth-
240    in-every-age-group-and-region-for-third-straight-year-significant-investments-still-needed-for-
241    court-and-facility-demand/ ;
242    https://www.thedinkpickleball.com/report-pickleball-participation-surges-to-
243    nearly-20-million-in-2024/ ;
244    https://www.pickleheads.com/guides/pickleball-statistics ;
245    https://www.custommarketinsights.com/report/pickleball-market/ ;
246    https://market.us/report/pickleball-market/
247  - Table tennis: https://en.wikipedia.org/wiki/International\_Table\_Tennis\_Federation
248  - Cricket: https://cricketrise.com/cricket-fans-in-india/ ;
249    https://www.nielsen.com/insights/2025/global-cricket-connecting-fans-driving-roi/
250  - Competitors & pricing: https://www.veo.com/en-us/pricing ; https://traceup.com/pricing

```

242       <https://www.pixellot.tv/> ; <https://www.hudl.com/pricing> ;  
243       <https://swing.vision/subscribe> ;  
244       <https://www.badproplusapp.com/>

## 6. user (2026-06-10T05:41:20.889Z)

```
1       # Next steps (live) ? owned-model + multisport
2
3       **Updated:** 2026-06-09 (end of the quality-iteration + multisport-recon session).
**Authoritative roadmap:**
4       `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md`. **Latest blow-by-blow:** memory `current-
state.md` / `session-handoff.md`.
5       This file = the prioritized "what to do next," refreshed each session.
6
7       ## The recommendation (one paragraph)
8       Build the **quality-first extraction harness** before anything else: a learned **TAL
head** (ActionFormer/ASFormer, MIT)
9       over trajectory features is the robust path ? it **decouples from the multi-court
"background games" confound** that the
10      unsupervised heuristic (and audio, and spatial court-gating) can't escape (all three
tested + failed; it's a *temporal*
11      rally-vs-non-rally problem). **Framework = ms-swift, base = Qwen3-VL** (Gemma-4
AMBER/bench) come **later** for the
12      conversational/grounding upgrade ? start with the head. **On-device is deferred**
(export-clean ops now so it stays
13      reachable). Everything stays **MIT/Apache-2.0/BSD** (code + data + weights). Bottleneck
= **data + per-sport trajectory
14      detectors**, not model architecture.
15
16      ## Latest findings (2026-06-09)
17      - **6-match badminton LOVO:** learned prototype **0.583** vs heuristic **0.480**
(learned leads since n=3, AUC 0.83?0.92,
18      decoupled from multi-court). The "contrast" metric (in-rally÷dead-time visibility)
predicts heuristic F1 monotonically.
19      - **?? Multisport (Roora pickleball + TT):** labels **ingest cleanly** (`curate convert-
cvat` ? 6 rallies/sport, sane).
20      **WASB-transfer SURPRISE (directional, n=6 each ? TINY):** the badminton shuttle
detector segments **TABLE TENNIS at
21      F1 0.909** out-of-the-box (!) but **pickleball only 0.308**. ? **The per-sport-
detector blocker is SPORT-DEPENDENT:** TT may
22      be servable with WASB short-term; pickleball needs a proper (MIT/Apache) ball
detector. Confirm/deny with more videos.
23      - **Owner field notes (2026-06-09):** camera height/distance **varied on purpose**
(great generalization data); **multi-court
24      adjacent-court SOUND dominates the signal** ? empirically reconfirms audio-foreground
isolation won't work + the
25      robustness-first learned-model path. Multi-court is the realistic (academy)
environment, so this *is* the target distribution.
26
27      ## Prioritized next steps
28      1. **[owner greenlight] Phase-0 (M0.x), $0/local:**
29      - **M0.4 ? Gen-0 TAL harness** (first-class): productionize `rally_seq_proto.py` ?
one-command train?eval?ship-gate** over
30      ActionFormer/ASFormer, reusing `metrics.match_intervals`. **Define the ship-gate
target** (explicit F1@tIoU; floor = beat
31      heuristic LOVO 0.480) ? "very high P/R" must be operationalized, not just "beat the
floor."
32      - **M0.1 ? corpus spine + the per-video event-index contract** (the conversational
seam: extract ? event store ? text-to-SQL).
33      - **M0.3 ? compliance:** purge Gemini-derived training labels
(`training.label_candidates`); WASB **C3** own-weights/clean-swap (gates ship).
34      2. **Multisport thread:**
35      - Ingest the Roora pickleball/TT rally CSVs into the corpus (collector `import-
local`).
```

```

36      - **Get more pickleball + TT videos** ? re-run the WASB-transfer eval to confirm/deny
TT?0.91, pickleball?0.31 (n=6 is too few).
37      - Once each sport has ?3 matches, run a **merged prototype LOVO across badminton + TT
(+ pickleball)** ? the "how would a
38      merged model do" experiment. Identify a clean MIT/Apache ball detector for sports
where WASB doesn't transfer (pickleball).
39      3. **Housekeeping:** **rename repo off "badminton" (multisport); confirm collector =
corpus/event-store system-of-record.
40
41      ## Pointers / artifacts
42      - Roadmap: `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` · Study:
`docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md`
43      (2026-06-09 banner) · Quality: `docs/archives/quality-iterations/2026-06-multivideo-
lovo/`.
44      - Eval corpus: `output/human_lovo_manifest.json` (badminton n=6). Roora CSVs:
`%TEMP%\roora\{pickleball,tabletennis}_rallies.csv`;
45      WASB-transfer trajectories: `%TEMP%\{pickleball,tabletennis}_traj.csv`. Code:
`backend/eval/{eval_partial_labels,rally_seq_proto,calibrate_local}.py`.
46      - ? No owned-model implementation until owner greenlights a milestone. Committed next
PLATFORM slice = async job model (single-tenant).
47

```

## 7. user (2026-06-10T05:41:37.890Z)

```

30      Priority: **P0** (do first) · **P1** · **P2** (lowest). Rows are grouped **Pending ? In-
flight ? Completed**, each sorted by priority. (C# IDs are stable identifiers ? they don't imply
order.)
31
32      | # | Workstream | Pri | Status | Where it stands | Next action / owner decision |
33      |---|---|---|---|---|---|
34      | | **? PENDING (needs action / not started)** | | | |
35      | C5 | **Output codec royalties** | **P0** | ? | Highlights are still encoded **H.264 +
AAC** (`libx264` in `backend/pipeline/stitcher/compiler.py`) ? encode-side AVC patent exposure
remains. | Switch output to **AV1 + Opus** (royalty-free) on an **Ada-class GPU (L4/L40)** for
HW encode. See §4. Decision: provision Ada-class GPU? |
36      | C9 | **Packaging / distribution** | **P0** | ? | No `pyproject.toml`; the diagnostic
`setup.py` shadows the build system. | Design in `DEFERRED_HARDENING_PLAN.md` (#7) ? **awaiting
your approval** before code. |
37      | C12 | **SaaS foundations (async/auth/tenancy)** | **P1** | ? | Sync `/api/process`
blocks on long videos; no auth/tenancy. | Multi-user/platform strategy in
`PLATFORM_ARCHITECTURE.md` (pluggable storage/compute, tenancy, orchestration); async slice =
`DEFERRED_HARDENING_PLAN.md` (#16). |
38      | | **? IN-FLIGHT (in progress / watch)** | | | |
39      | C4 | **Runtime AI strategy (A/B/C + owned model)** | **P0** | ? | Live = **C) Gemini
2.5 Flash, paid tier**. Long-term = **own a self-hosted generative video-QA model** (fine-tuned
open VLM) as primary, **paid Gemini always-ready as fallback** via the provider registry. | Make
"provider registry + paid fallback" a core principle; build the owned model **after** the
scaffolding/flywheel are solid. See `PLATFORM_ARCHITECTURE.md` §5A. |
40      | C13 | **Beachhead validation** | **P0** | ? | Market research
(`PMF_MARKET_RESEARCH.md`) recommends **India badminton academies** as the #1 beachhead.
Unvalidated. | Cheapest tests: 10?15 academy/coach discovery interviews; rally-detection P/R
benchmark on 100+ real amateur clips; fake-door pricing (B2C ~$99/yr, B2B academy ~$600/yr). See
§6. |
41      | C8 | **Golden-set vendoring spend** | **P1** | ? | ADR-006: **Roora (Bengaluru)**
selected for an initial paid pilot; licensed/owned video only (no user uploads to vendors). |
Confirm quality-per-dollar from the pilot; then GS-3 `HumanVendorProvider`. See
`GOLDEN_SET_VENDORING.md`. |
42      | C3 | **Shuttle-tracking weights provenance** | **P1** | ? ?? | WASB-SBDT **code is MIT
(clear)**; distributed **checkpoints are academic-data-trained** ? not covered by the MIT code
grant. MonoTrack/YOLO-NAS banned (noncommercial). | Train-own weights (Phase-4 roadmap) **or**
confirm per-checkpoint terms ?? before commercial deploy. |
43      | C14 | **Owned-model base licensing + training-data policy** | **P1** | ? ?? | Owned
video-QA model (PLATFORM §5A) must build on **Apache-2.0-clean, video-capable** bases ?
**Qwen2.5-VL-7B/32B** (Apache; 3B/72B are *not*), **Qwen3-VL-4B/8B**, or **Gemma 4** ? to keep
weights private, gate, and monetize. **Reject Gemma 3** (viral derivative clause ? permanent

```

Google PUP), Llama Vision (naming + 700M-MAU), `gpt-oss` (text-only). **\*\*Hard rule: never train on paid-LLM (Gemini/OpenAI/Anthropic) outputs\*\*** (competing-model clause). The moat is the **\*\*consented dataset + eval harness\*\***, not the model. | Lock to Apache-2.0; **\*\*open-model teachers only\*\***; separate explicit **\*\*training opt-in\*\***; **\*\*exclude minors\*\*** from the training pool; per-clip consent ledger. See PLATFORM §5A/§7. |

44 | C6 | **\*\*Inbound HEVC decode\*\*** | **\*\*P2\*\*** | ? ?? | Decoding users' GoPro H.264/**\*\*HEVC\*\*** uploads is unavoidable; HEVC pools are fragmented + highest-uncertainty. | Low/uncertain residual; confirm decode-side exposure with counsel before launch. |

45 | C7 | **\*\*Hosting unit-economics\*\*** | **\*\*P2\*\*** | ? | Per-video cost is *\*cheap\** for Gemini (~\$0.05 input for a 10-min clip); risk is volume × latency, not unit price. Self-host/offline costs unmeasured. | **\*\*Benchmark \$/video & latency on the target VM\*\*** for A/B/C before pricing. See §5. |

46 | C10 | **\*\*Value layer (product surface)\*\*** | **\*\*P2\*\*** | ? | Minimal exports shipped (EDL/JSON/CSV + `basic\_stats`, PR #51). | Deepen: per-rally tags/notes that feed golden, per-player/match stats, clip gallery. See §6. |

47 | C11 | **\*\*Input-quality enforcement\*\*** | **\*\*P2\*\*** | ? | Per-video observability reports surface a customer verdict + footgun flags incl. audio-present (PR #50). | Turn more `VIDEO\_RECORDING\_GUIDELINES` dimensions into validators/report rules. |

48 | C2 | **\*\*supervision` ByteTrack pin\*\*** | **\*\*P2\*\*** | ? | Pinned `<0.30.0` (deprecation ahead). | Migrate before lifting the pin (BACKLOG). Not a legal risk, a maintenance one. |

49 | | **\*\*? COMPLETED\*\*** | | | |

50 | C1 | **\*\*AGPL / copyleft removal\*\*** | ? | ? | ADR-005 shipped: AGPL `ultralytics` YOLOv8 replaced with **\*\*torchvision Faster R-CNN (BSD) + `supervision` ByteTrack (MIT)\*\***. No AGPL code/weights in the stack. | Done. Keep new deps permissive. |

51

52 ---

53

54 ## 2. Licensing & dependency posture (refreshed risk table)

55

56 Verdicts updated against the current stack (`requirements.txt` + code). Original

57 2026-05-30 verdicts and the 25-claim verification transcript are in

58 `archives/MONETIZATION\_AUDIT.md`.

59

| Dependency                                                                                 | License              | SaaS obligation?           | Current verdict      | Notes                                                                                  |
|--------------------------------------------------------------------------------------------|----------------------|----------------------------|----------------------|----------------------------------------------------------------------------------------|
| --- --- --- --- ---                                                                        |                      |                            |                      |                                                                                        |
| ~~`ultralytics` + `yolov8n.pt`~~                                                           | AGPL-3.0             | Yes (§13)                  | ? <b>**REMOVED**</b> | Replaced per ADR-005 ? no longer in the stack.                                         |
| <b>**torchvision**</b> detectors (Faster R-CNN/RetinaNet)                                  | BSD-3                | No                         | ? LIVE               | The chosen replacement; person = COCO class (see `yolo_hybrid.py` `_PERSON_CLASS_ID`). |
| <b>**`supervision`**</b> (ByteTrack)                                                       | MIT                  | No                         | ? LIVE               | Pinned `<0.30.0` ? migrate before bump (C2).                                           |
| <b>**WASB-SBDT**</b> (shuttle)                                                             | MIT (code)           | No                         | ? code clear         | Weights provenance open (C3).                                                          |
| <b>**MonoTrack / YOLO-NAS**</b>                                                            | noncommercial        | ? ?                        | BANNED               | Do not use.                                                                            |
| <b>**Gemini API**</b> (paid tier)                                                          | service terms        | Paid: no training/review   | ? LIVE WATCH         | Must bill the paid tier to keep user video private (C4).                               |
| H.264 / HEVC / AAC codecs                                                                  | patent pools         | royalties on encode/decode | ? ??                 | Separate from ffmpeg's license. Encode-side open (C5); decode-side residual (C6).      |
| ffmpeg/ffprobe, google-genai, torch, opencv, numpy, fastapi/uvicorn/pydantic/jinja2/dotenv | LGPL/Apache/BSD/MIT  | No                         | ? CLEAR              | Invoked as subprocess / imported; all permissive.                                      |
| <b>**`obsreport`**</b> (reports)                                                           | permissive (private) | No                         | ? soft dep           | Pin the `git+ssh@vX` tag once the repo is published (BACKLOG).                         |

71

72 **\*\*Bottom line:\*\*** the two hard blockers from the audit (Ultralytics; banned weights)

73 are resolved/avoided. Remaining commercial-legal items are **\*\*codec royalties (C5/C6)\*\***

74 and **\*\*weights provenance (C3)\*\***, plus the metered-API privacy posture (C4).

75

76 ---

77

78 ## 3. Runtime AI strategy ? A/B/C

79

80 The semantic rally-boundary step is the one external/metered dependency. Three paths

81 (from the audit), with current standing:

82

```

83 - *(A) Eliminate ? offline learned segmenter.** ActionFormer (MIT) / our own learned
84 classifier on `candidate_segments.stats`. **Best steady-state margin** (no per-call
85 cost, no external dep) and matches the Phase-4 roadmap (substrate = WASB
86 shuttle-motion, decided in ADR-004). **Most engineering.** ? *target*
87 - *(B) Self-host VLM.** Qwen2.5-VL-7B/32B (Apache-2.0) or InternVL2.5-8B (MIT). No
88 per-call fees; pays in GPU VRAM (~16?24 GB for 7B). ?? Qwen 3B is noncommercial.
89 - *(C) Keep Gemini 2.5 Flash (paid).** Fastest, cheap per video, video-native;
90 ongoing metered cost + external dependency. **? current live path.**
91
92 Hosted alternatives for (C) comparison: Anthropic Claude (no training on
93 inputs/outputs; customer owns outputs; **no native video** ? weaker here);
94 OpenAI (no training by default; ZDR available); Gemini remains strongest video-native.
95
96 ---
97
98 ## 4. Codec / output economics (the live gap ? C5)
99
100 **The trap:** ffmpeg being free (LGPL) does **not** grant the *patent* rights to the
101 codecs it implements. Royalties attach to the act of encode/decode.
102
103 - **Encode side (we control it):** highlights currently emit **H.264/AAC**. AVC (Via LA)
104 is first 100k units/yr royalty-free with a permanent "free to end-users over the
105 internet" provision ? a small SaaS likely falls under thresholds, but ?? confirm.
106 - **Mitigation:** emit **AV1 (SVT-AV1, BSD + AOMedia royalty-free patent grant) + Opus
107 (royalty-free)**. NVENC AV1 HW-encode needs **Ada-class GPUs (L4/L40)**; Ampere/Turing
108 must use **SVT-AV1 (CPU)**.
109 - **Decode side (unavoidable, C6):** inbound GoPro **HEVC** decode ? fragmented pools,
110 highest uncertainty; low residual.
111
112 **Recommendation:** provision an **Ada-class GPU (L4/L40)** and standardize highlight
113 output on **AV1 + Opus** ? removes the encode-side royalty question almost entirely.
114 *Implementation lives in `backend/pipeline/stitcher/compiler.py` (codec args).*
115
116 ---
117
118 ## 5. Hosting unit-economics (C7)
119
120 Reference workload: ~30-min 1080p match on a cloud NVIDIA GPU VM.
121
122 - *(C) Gemini 2.5 Flash (paid):** input $0.30/1M tokens, output $2.50/1M; video ~263
123 tok/s ? 10-min clip ? 158k input tokens ? **~$0.05 input**. Cheap per unit; the
124 economic risk is **volume × latency**. Flash-Lite is cheaper; Pro ~10×.
125 - *(B) Self-host Qwen2.5-VL-7B:** $0 API; cost = GPU-hours / throughput; needs ?24 GB
126 VRAM (L4/A10). $/video vs Gemini is **open ? measure**.
127 - *(A) Offline ActionFormer/learned:** cheapest at inference; cost = one-time training
128 + light GPU. **Best steady-state margin.**
129
130 > The A/B/C $/video comparison depends on GPU instance + batch throughput ?
131 > **benchmark on the target VM before pricing the product.**
132
133 ---
134
135 ## 6. Product-market fit & the quality/data phase
136
137 ### Beachhead (from market research, 2026-06-07 ? see
138 [`PMF_MARKET_RESEARCH.md`](PMF_MARKET_RESEARCH.md))
139 The static-camera constraint is a **B2B filter**: the buyer must already record from a
140 fixed angle, which points away from the median casual player and toward coaches/clubs.
141 Ranked beachheads:
142 1. **Badminton coaching academies & serious players, India** *(strongest)* ? 20M+
143 players,
144 formalizing academy ecosystem (Khelo India), genuine fixed-angle recording, coaches
145 who already monetize, badminton whitespace vs incumbents. Sell to the **academy**,
146 not
147 the individual; needs INR-tuned pricing.

```

145 2. **\*\*US/global pickleball players & clubs\*\*** ? fastest-growing sport (19.8M US, +45.8% YoY)  
 146 but **\*\*SwingVision already owns single-camera tennis/pickleball AI\*\*** (USD 179.99/yr) ?  
 147 head-to-head, not greenfield.  
 148 3. **\*\*Badminton academies/clubs in East/SE Asia + Denmark\*\*** ? highest density; China  
 needs  
 149 local hosting/partners; Denmark/EU small but high-WTP, easy to pilot.  
 150  
 151 Pricing anchors **\*(estimate)\***: software-only BYO-camera **\*\*B2C USD 60?180/yr\*\***, **\*\*B2B**  
 152 academy/club USD 300?1,500/yr\*\* per location (undercuts hardware-bundled Veo/Pixellot/  
 153 Spiideo; comparable to SwingVision software-only).  
 154

## 8. user (2026-06-10T05:41:38.555Z)

1 # Owned-model implementation plan (roadmap) ? DRAFT for owner review  
 2  
 3 **\*\*Status:\*\*** PLAN / roadmap. No owned-model **\*implementation\*** starts until the owner  
 greenlights a specific  
 4 milestone. Paid Gemini stays the always-ready inference/fallback throughout (never a  
 training source).  
 5 **\*\*Grounded in:\*\*** `docs/archives/research/OWNED\_MODEL\_TRAINING\_STUDY.md`, the n=6 LOVO +  
 learned prototype  
 6 (`docs/archives/quality-iterations/2026-06-multivideo-lovo/`), and a verified strategy  
 workflow (2026-06-09).  
 7  
 8 **## Design principle (owner directive):** **\*envision now, build later\***  
 9 Every deferred piece (conversational-QA, on-device, multisport, the VLM) must be  
**\*\*architecturally envisioned**  
 10 **now\*\*** so we never hit a mammoth refactor. We **\*\*design the seams\*\*** (data contracts,  
 export boundaries, the  
 11 event store) up front and **\*\*implement behind them\*\*** when each milestone is greenlit.  
 Punting a **\*feature\*** is fine;  
 12 punting its **\*seam\*** is not.  
 13  
 14 **## North Star (reframed per PR #61 review)**  
 15 - **\*\*Immediate deliverable:\*\*** a **\*\*training framework + harness\*\*** that produces a model  
 with **\*\*very high precision/**  
 16 recall for extracting highlights + rallies from a clip\*\*. **\*Quality is the first-order**  
 objective.\*  
 17 - **\*\*Eventual product:\*\*** a **\*\*single, sport-agnostic, conversational\*\*** video model ?  
 upload a clip, then ask  
 18 contextual questions ("longest rally", "make a clip of all service faults", "rallies  
 over 20 shots").  
 19 - **\*\*On-device is LATER\*\*** (a model-compression goal so power users save upload  
 cost/hassle) ? definitely on the  
 20 roadmap, but **\*\*not\*\*** the immediate target. Don't over-index the strategy on it now.  
 21 - **\*\*The moat = the consented dataset + the eval harness, not the weights.\*\***  
 22  
 23 **## Hard guardrails (non-negotiable ? "never corrupted")**  
 24 - **\*\*Commercial-clean licensing for EVERY dependency ? code, data, AND model weights: MIT**  
 / Apache-2.0 / BSD only.\*\*  
 25 (Ratifies the prior "Apache-2.0 only" to include MIT/BSD.) Reject viral/NC/custom-  
 restrictive: Gemma 3 (viral),  
 26 **\*\*Gemma 3n\*\*** (Gemma Terms follow derivatives ? even cleaner reject), Llama 4 (700M-MAU  
 cap), Commons-Clause repos.  
 27 - **\*\*Never train on paid-LLM (Gemini/OpenAI/Anthropic) outputs.\*\*** Also forbidden: public  
 grounding-CoT datasets that  
 28 were Gemini-generated (e.g. TVG-R1's 56K) ? we must originate our own CoT supervision  
 (a real, binding cost).  
 29 - **\*\*Exclude minors; per-user training data needs a separate explicit opt-in; split data**  
 by match.\*\*  
 30 - ? **\*\*Base-model pretraining provenance is unauditabile for \*every\* open VLM (incl.**  
 Qwen3-VL). **\*\* Our "no paid-LLM**  
 31 **training" rule is enforceable at \*our fine-tuning data\* layer, not the base's**

```

pretraining ? an **explicit owner
32     risk-acceptance** is required to use any open VLM. (Governed by the base's license,
not our pipeline.)
33
34     ## The model strategy (verified 2026-06-09)
35     **Framework ? base** (the Gemma-4-vs-ms-swift clarification): a *framework* is the forge
(run SFT/GRPO, emits
36     weights); a *base* is the metal you forge. You pick one framework and feed it any clean
base.
37     - **Framework: ms-swift (Apache-2.0)** ? the one forge that does native **video + audio
+ timestamp-grounding +
38     GRPO/RFT with a custom video reward** in one stack. Keep **unsloth** (Apache-2.0 core;
*avoid its AGPL Studio UI*)
39     as the 8 GB-local SFT/compression tool for the deferred on-device goal. ? Point-in-
time snapshot ? re-verify at impl.
40     - **Base (when we reach the VLM): Qwen3-VL (Apache-2.0, uniform across sizes)** ? long-
video context, purpose-built
41     **timestamp emission** (second-level localization), multi-turn video chat. Re-confirm
the exact size's LICENSE
42     pre-release. **Gemma-4 stays AMBER and bench-only** ? its Apache grant is likely but
its Prohibited-Use posture is
43     unconfirmed (counsel needed), *and* its video/timestamp capability is disputed; not
the primary extractor.
44     **InternVL3 (MIT/Apache)** = clean fallback if a Qwen blocker appears. ? base swap is
cheap *only within
45     timestamp-capable Apache bases* (timestamp data-format + grounding-CoT are base-
specific).
46     - **START WITH THE TAL HEAD, NOT THE VLM** (the decisive call): fine-tuned VLMs **miss
strict temporal boundaries**
47     (best RL-post-trained video VLM ? R@0.7 50% on open-domain benchmarks; "coarse
regions, not accurate boundaries").
48     **Boundary accuracy IS the product.** The head (`rally_seq_proto.py`) is already de-
risked on our corpus (held-out
49     AUC 0.83?0.92), trains in minutes at ~4.5 GB on the 2070 Super ($0), is ~1M params
(trivially compressible later),
50     and **doubles as a pseudo-label teacher + the honest baseline the VLM must beat.** So:
**head now ? VLM later**,
51     strictly sequenced. When the VLM comes, prefer **two-stage** (VLM coarse-proposes,
head refines boundaries) over
52     wholesale replacement. *(Open: the VLM-ceiling number is open-domain; our amateur
footage may differ ? re-measure.)*
53
54     ## Conversational video-QA ? build the bookkeeping NOW, punt the model
55     **Punt the conversational *feature*** until extraction quality is solid (current LOVO
~0.583 ? "very high"; a chat UI
56     over a weak extractor exposes the weakness). **Reject end-to-end-VLM-answers-each-turn**
(expensive; bad at exact
57     counting; the strong teachers are license-forbidden). **But build the *seam* now** (the
brief requires it):
58     - **Architecture = "extract once, query many":** EXTRACTION (TAL head ? later VLM) emits
structured timestamped
59     spans ? a durable **per-video EVENT STORE** ? a **structured QUERY layer** (text-to-
SQL + ffmpeg clip-cut). The
60     cited queries ("longest rally", "all service faults", "rallies > 20 shots") are
**structured-numeric ? SQL, not a
61     VLM job** (SQL counts exactly where VLMs hallucinate).
62     - **Designed-now seam:** make the **per-video event-index schema a first-class data
contract** alongside the M0.1
63     corpus spine: per rally `{video_id(MD5), rally_ordinal, sport, start, end, shot_count,
ending_reason/fault_tags,
64     derived_clip_ref, provenance, confidence}`. **Reserve the event/fault-tag taxonomy
(e.g. `service_fault`) up front**
65     even if unpopulated. ? This is a real DB delta (a `PRAGMA user_version` migration):
`play_segments` today has
66     `video_id/start_time/end_time/duration/server/receiver/metadata` only; `highlights`

```

isn't a table yet ? so it's an

67       \*extension with new columns + a highlights table\*, not a no-op. \*\*Punt\*\* the NL  
parser/agent until quality clears the bar.

68       - Paid Gemini may answer genuinely open-ended edge questions \*\*at inference\*\* over our  
structured index (allowed ?

69       it reasons over our data, it is not trained on its outputs).

70

## 9. assistant (2026-06-10T05:41:50.309Z)

Internal citations check out so far. Now verifying the load-bearing external claims with web searches.

## 10. user (2026-06-10T05:41:56.078Z)

Web search results for query: "Machaxi funding Rainmatter Prakash Padukone badminton \$1.5 million"

Links: [{"title": "Machaxi Secures \$1.5 Million to Expand AI-Powered Badminton Coaching Across India", "url": "https://india.entrepreneur.com/news-and-trends/machaxi-secures-15-million-to-expand-ai-powered-badminton/493070"}, {"title": "Machaxi Raises \$1.5 Million to Scale AI-Powered Badminton Coaching Across India ? Outlook Business", "url": "https://www.outlookbusiness.com/start-up/investors/machaxi-raises-15-million-to-scale-ai-powered-badminton-coaching-across-india"}, {"title": "Machaxi aces \$1.5m fund raise from Padukone and Nithin Kamath?s Rainmatter | 1 Indian Television Dot Com", "url": "https://www.indiantelevision.com/mam/digital/machaxi-aces-\$1.5m-fund--raise-from-padukone-and-nithin-kamath%E2%80%99s-rainmatter-250611"}, {"title": "Machaxi raises \$1.5 Mn from Rainmatter, Prakash Padukone, others", "url": "https://entrackr.com/snippets/machaxi-raises-15-mn-from-rainmatter-prakash-padukone-others-9353316"}, {"title": "Machaxi raises \$1.5mn from Prakash Padukone, Rainmatter and others - MediaBrief", "url": "https://mediabrief.com/machaxi-raises-1-5m/"}, {"title": "Zerodha's Rainmatter, others invests \$1.5 million in sportstech startup Machaxi", "url": "https://indianstartupnews.com/funding/zerodhas-rainmatter-invests-in-sportstech-startup-machaxi-9353417"}, {"title": "Machaxi Raises \$1.5M to Revolutionize Grassroots Badminton with AI and Prakash Padukone", "url": "https://www.tice.news/tice-dispatch/filling-the-sports-gap-rainmatter-funds-ai-sports-startup-machaxi-9353716"}, {"title": "Machaxi Raises \$1.5M from Prakash Padukone, Rainmatter and Others to Expand AI-Based Grassroots Badminton Training - Passionate In Marketing Machaxi Raises \$1.5M to Expand AI-Powered Grassroots Badminton Coaching Across India", "url": "https://www.passionateinmarketing.com/machaxi-raises-1-5m-from-prakash-padukone-rainmatter-and-others-to-expand-ai-based-grassroots-badminton-training/"}]

Based on the search results, here's what I found:

Machaxi, a Bengaluru-based sports-tech startup focused on grassroots coaching, has raised \$1.5 million in funding to scale its AI-powered training model and expand operations into three more cities?Hyderabad, Pune, and Chennai.

The funding round was led by Rainmatter, the investment arm of Zerodha, and included participation from Indian badminton legend Prakash Padukone as well as existing backers.

Under the "Machaxi x Padukone School of Badminton" initiative, the company aims to establish more than 1,000 coaching centres across India within the next four years. A cornerstone of the partnership is an AI?based coaching system designed to standardise training quality, assist in performance tracking and ensure scalability without replacing human coaches.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

## 11. user (2026-06-10T05:41:57.534Z)

Web search results for query: "SwingVision Wefunder revenue 2024 paying subscribers ARR"

Links: [{"title": "SwingVision on Wefunder 2024 - Kingscrowd", "url": "https://kingscrowd.com/swingvision-on-wefunder-2024/"}, {"title": "SwingVision



on Wefunder 2025 - Kingscrowd", "url": "https://kingscrowd.com/swingvision-on-wefunder-2025/"}], {"title": "SwingVision Stock Price, Funding, Valuation, Revenue & Financial Statements", "url": "https://www.cbinsights.com/company/swingvision/financials"}, {"title": "SwingVision : Automated Stats, Highlights & Officiating Using Just Your Phone | Wefunder, Home of the Community Round", "url": "https://wefunder.com/swingvisioncf2025"}, {"title": "WeFunder Stock Price, Funding, Valuation, Revenue & Financial Statements", "url": "https://www.cbinsights.com/company/wefunder/financials"}, {"title": "Wefunder Updates, Team, and Funding Progress | Wefunder, Home of the Community Round", "url": "https://wefunder.com/wefunder"}, {"title": "SwingVision Scores \$6 Million Series A to Bring AI to Athletes", "url": "https://www.prweb.com/releases/swingvision-scores-6-million-series-a-to-bring-ai-to-athletes-301960272.html"}, {"title": "SwingVision Pro Plans & Pricing | Tennis & Pickleball App", "url": "https://swing.vision/subscribe"}]

Based on the search results, here's what I found about SwingVision's 2024 metrics:

**\*\*Paying Subscribers (2024):\*\*** By the end of 2024, SwingVision reported more than 20,000 paying subscribers on its platform.

**\*\*Revenue (2024):\*\*** In its most recent fiscal year (2024), SwingVision recorded \$2,746,593 in revenue, a year-over-year increase of about 12% from \$2,455,089 in 2023.

**\*\*Annual Recurring Revenue (ARR):\*\*** The platform has more than 20,000 paying subscribers, 100 D1-D3 college teams, more than \$4 million in annual recurring revenue, and has officiated over 500 USTA matches with electronic line calling.

These figures come from SwingVision's 2024/2025 Wefunder campaign information. The company has shown strong growth, with subscriber numbers increasing from 15,000 in mid-2024 to over 20,000 by year-end, and ARR growing from \$2.5 million to over \$4 million.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

## 12. user (2026-06-10T05:41:59.259Z)

Web search results for query: "VISIST.AI badminton Padukone School of Badminton skill assessment Bengaluru academies"

Links: [{"title": "VISIST.AI partners with Padukone School of Badminton to digitise athlete skill assessments", "url": "https://theprint.in/india/visist-ai-partners-with-padukone-school-of-badminton-to-digitise-athlete-skill-assessments/2716417/"}, {"title": "VISIST.AI Partners with Padukone School of Badminton to Revolutionize Athlete Training | Sports-Games", "url": "https://www.devdiscourse.com/article/sports-games/3533385-visistai-partners-with-padukone-school-of-badminton-to-revolutionize-athlete-training"}, {"title": "Indian badminton players use artificial intelligence to up their game - Mugglehead Investment Magazine", "url": "https://mugglehead.com/indian-badminton-players-use-artificial-intelligence-to-up-their-game/"}, {"title": "Ananya Verma - AI Enthusiast | Exploring the world of Intelligent Technology | LinkedIn", "url": "https://www.linkedin.com/in/ananya-verma-6339b1251/"}, {"title": "Padukone School of Badminton", "url": "https://karthiks-testing-site.webflow.io/"}, {"title": "Bangalore Badminton Courts ? Padukone School Of Badminton", "url": "https://padukoneschoolofbadminton.com/pages/bangalore-badminton-courts"}, {"title": "Padukone School of Badminton Expands to 75 Centres in First Year, Targets 250 by 2027 | Entrepreneur", "url": "https://www.entrepreneur.com/en-in/news-and-trends/padukone-school-of-badminton-expands-to-75-centres-in-first/493030"}, {"title": "Padukone School Of Badminton", "url": "https://padukoneschoolofbadminton.com/"}, {"title": "About Padukone School of Badminton - Vision & Mission ? Padukone School Of Badminton", "url": "https://padukoneschoolofbadminton.com/pages/about-padukone-school-of-badminton"}, {"title": "PPSM", "url": "http://www.ppsm.org.in/children-coaching-programs.html"}]

Based on the search results, here's what I found about VISIST.AI, badminton, and the Padukone School of Badminton:

## Partnership Overview

VISIST.AI, a sports tech start-up incubated by the FSID at the Indian Institute of Science, announced a tie-up with the Padukone School of Badminton to digitise skill assessments across the academy's network.

## VISIST.AI Technology

VISIST.AI is leveraging artificial intelligence to deliver performance insights, skill assessments, and data-driven coaching for athletes and sports academies. More specifically, VISIST.AI's program will inform the player about flaws in their posture, grip, swing velocity and overall technique.

## Implementation and Scale

Through this partnership, VISIST.AI's Visual Intelligence platform will be deployed across PSB's academy ecosystem to enable data-driven athlete evaluations, personalised feedback, and evidence-based training insights. More than 70 academies in the city have been employing the technology, many of which are overseen by the Padukone School of Badminton.

## Bengaluru Academies

Padukone School of Badminton offers training in badminton courts of Bengaluru, Karnataka with specially designed training programs for all kids, professional and adults, providing world-class facilities and learning from India's top coaches.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

### 13. user (2026-06-10T05:42:20.995Z)

Web search results for query: "PB Vision pickleball pricing API "\$8" per hour video"

Links: [{"title": "PB Vision API Partner Guide - PB Vision", "url": "https://help.pb.vision/articles/2793895-pb-vision-api-partner-guide"}, {"title": "Get Your DUPR Rating with AI & PB Vision - thePicklebase | ThePickleBase", "url": "https://www.thepicklebase.com/coaching-tips/future-pickleball-ratings-getting-dupr-rating-power-ai"}]

Based on the search results, I found information about PB Vision's API pricing:

For the PB Vision API, HD video (1080p or lower) costs \$8 per hour of video processed, which works out to about \$1.60 per 12-minute game. 4K video (which must be explicitly enabled) costs \$12 per hour, or about \$2.40 per 12-minute game.

The PB Vision API provides transcoding, analysis, and analytics data delivery in one automated pipeline for businesses that generate or manage pickleball video. The service offers flexible billing models, including pre-purchasing video-processing minutes or invoice-based usage for high-volume partners.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

### 14. user (2026-06-10T05:42:22.465Z)

Web search results for query: "Stupa Sports Analytics funding BWF badminton instant review table tennis"

Links: [{"title": "Stupa wins BWF approval for badminton review system", "url": "https://itbrief.in/story/stupa-wins-bwf-approval-for-badminton-review-system"}, {"title": "Badminton, Padel, Table Tennis to use AI driven IRS after BWF Approval: WTT, India Open set for New Review Procedure - myKhel", "url": "https://www.mykhel.com/more-sports/badminton-padel-table-tennis-to-use-ai-driven-irs-after-bwf-approval-wtt-india-open-set-"}]

for-new-420431.html"}, {"title": "Stupa Sports? AI-Powered Instant Review System achieves official BWF certification for global Badminton tournaments - APN News | Authentic Press Network News", "url": "https://www.apnnews.com/stupa-sports-ai-powered-instant-review-system-achieves-official-bwf-certification-for-global-badminton-tournaments/"}, {"title": "Stupa Sports? AI review system gets BWF certification for global tournaments", "url": "https://mediabrief.com/stupa-sports-ai-review-system-gets-bwf-certification-for-global-tournaments/"}, {"title": "Stupa Sports' AI Review System Is Now BWF Certified for Global Badminton", "url": "https://technosports.co.in/stupa-sports-ai-review-is-now-bwf-certified/"}, {"title": "Stupa Sports Analytics", "url": "https://stupasports.ai/"}, {"title": "Stupa Sports Analytics Nets INR 28 Cr To Offer AI Solutions To Sports Federations", "url": "https://inc42.com/buzz/stupa-sports-analytics-nets-inr-28-cr-to-offer-ai-solutions-to-sports-federations/"}, {"title": "Stupa Sports Analytics Secures Rs 280 Million in Pre-Series A Funding Round ? Outlook Business", "url": "https://www.outlookbusiness.com/corporate/stupa-sports-analytics-secures-rs-280-million-in-pre-series-a-funding-round"}]

Based on the search results, here's what I found about Stupa Sports Analytics:

## Funding

Stupa Sports Analytics was founded in 2020 by Megha Gambhir and Deepak Malik. The company has secured significant funding: before its Pre-Series A round, Stupa had raised Rs 75 million in seed and pre-seed funding, and the company raised INR 28 Cr to expand into US, UK, Australia and the Middle East markets.

## BWF Badminton Instant Review System

Stupa Sports received certification from the Badminton World Federation for its Instant Review System for use in sanctioned tournaments, allowing the Gurugram-based sports technology company's AI-based line-calling and review product to be used across BWF events. BWF testing found the system achieved 99% accuracy, above the 98% threshold required for approval.

## Table Tennis and Multi-Sport Presence

Recent commercial work has included deployments linked to the Generali Hexagon Cup padel tournament, the Pickleball Champions League Asia Finals, Major League Table Tennis and the Prime Volleyball League. The company provides fan engaging content for TV broadcasters, majorly catering to Tier 2 sports like badminton, padel, pickleball, volleyball and rugby, among others.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

### 15. user (2026-06-10T05:42:22.804Z)

Web search results for query: "Game Theory sports startup Bengaluru smart courts highlights Gopichand Rainmatter"

Links: [{"title": "GameTheory - 2026 Company Profile, Team, Funding, Competitors & Financials - Tracxn", "url": "https://tracxn.com/d/companies/gametheory/\_\_\_pd0B46xo2J1QH-rpu1vpiw-gl-kIJXu3VNWhc9otULM"}, {"title": "Using computer vision, Game Theory helps sports players ...", "url": "https://yourstory.com/2024/02/computer-vision-game-theory-sports-tech-startup-nithin-kamath-rainmatter"}, {"title": "Sportstech Startup Game Theory Bags Funding From Nithin Kamath, Rohan Bopanna", "url": "https://inc42.com/buzz/game-theory-bags-funding-from-nithin-kamath-rohan-bopanna-to-provide-sports-technology/"}, {"title": "Sports Tech Startup Game Theory Secures Funding from Badminton Legend Pullela Gopichand and Others | Startup Story", "url": "https://startupstorymedia.com/insights-sports-tech-startup-game-theory-secures-funding-from-badminton-legend-pullela-gopichand-and-others/"}, {"title": "Game Theory Ropes In Chief Badminton Coach Pullela Gopichand, Others As Investors", "url": "https://inc42.com/buzz/sportstech-startup-game-theory-ropes-in-chief-badminton-coach-pullela-gopichand-others-as-investors/"}, {"title": "Bengaluru-Based Game Theory Raises Funding From Several Prominent Sports Personalities - Vetted Sports", "url": "https://vettedsports.com/bengaluru-based-game-theory-raises-funding-from-several-"}]

```
prominent-sports-personalities/"},"title":"Bengaluru-based Game Theory raises funding from prominent sports personalities","url":"https://indianstartupnews.com/funding/bengaluru-based-game-theory-raises-funding-from-prominent-sports-personalities-6710948"}]
```

Based on the search results, here are the key highlights about Game Theory:

## Company Overview

Game Theory is a seed company based in Bengaluru, founded in 2019 by Sudeep Kulkarni. The company converts facilities into smart courts and gamifies the experience of playing sports. It uses smart courts to track scores, steps, heatmaps, and strokes.

## Funding and Investors

Game Theory has raised \$3.13M in funding from investors like Techstars, Zerodha and Rainmatter. The company picked up \$2 million in a seed funding round in October last year led by Rainmatter Capital.

More recently, Game Theory announced the successful closure of its 'Masters' funding round, receiving substantial financial investments from Indian sporting legends, including Padma Bhushan awardee and chief national badminton coach, Pullela Gopichand. Other notable investors in this round include former professional squash player Saurav Ghosal and renowned table tennis player Sharath Kamal.

## Smart Court Technology

Game Theory's technology stack leverages computer vision-driven analytics, artificial intelligence-powered performance tracking, and real-time feedback to help athletes develop from foundational skills to elite competitive levels.

## Current Operations

The company has 15 venues in Bengaluru, with five more expected to open soon.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

16. user (2026-06-10T05:42:39.287Z)

## Gemini API Pricing Summary

### Gemini 2.5 Flash (Standard)

- **Input:** "\$0.30 (text / image / video)" and "\$1.00 (audio)"
- **Output:** "\$2.50"

### Gemini 2.5 Flash-Lite (Standard)

- **Input:** "\$0.10 (text / image / video)" and "\$0.30 (audio)"

### Gemini 3.5 Flash (Standard)

- **Input:** "\$1.50" per 1M tokens
- **Output:** "\$9.00" per 1M tokens (including thinking tokens)

## Batch Pricing

Yes, batch pricing is approximately half the standard rate. For example, Gemini 2.5 Flash batch shows "\$0.15 (text / image / video)" input and "\$1.25" output?exactly 50% of the standard tier prices.

17. user (2026-06-10T05:42:42.079Z)

Web search results for query: "SwingVision badminton support roadmap 2025 2026 racket sports expansion padel"

Links: [{"title":"SwingVision SWOT Analysis: Ready-Made ?

```
CanvasBusinessModel.com", "url": "https://canvasbusinessmodel.com/products/swingvision-swot-analysis"}, {"title": "SwingVision Raises $6M to Reach Pickleball and Padel - Profluence", "url": "https://profluence.com/swingvision-raises-6m-to-reach-pickleball-and-padel/"}, {"title": "SwingVision FAQ | Tennis & Pickleball App Support", "url": "https://swing.vision/faq"}, {"title": "SwingVision - YouTube", "url": "https://www.youtube.com/@SwingVisionApp"}, {"title": "AI tennis app SwingVision raises US$6m to expand into ...", "url": "https://www.sportspro.com/news/swingvision-ai-tennis-app-pickleball-padel/"}, {"title": "SwingVision: Tennis Pick... ? Review 2026: Sentiment & Intel", "url": "https://marlvel.ai/intel-report/sports/com-mangolytics-swing"}, {"title": "Racquet Sports Equipment Market Size, Share | Growth [2034]", "url": "https://www.fortunebusinessinsights.com/racquet-sports-equipment-market-115793"}, {"title": "Padel Racket Market Trends, Share and Forecast, 2026-2033", "url": "https://www.coherentmarketinsights.com/industry-reports/padel-racket-market"}, {"title": "SwingVision Scores $6 Million Series A to Bring AI to Athletes", "url": "https://www.prweb.com/releases/swingvision-scores-6-million-series-a-to-bring-ai-to-athletes-301960272.html"}]
```

Based on the search results, here's what I found about SwingVision's badminton support and racket sports expansion plans:

## Badminton Support

While the search results mention badminton in the context of SwingVision's target market, there is no specific information about a dedicated badminton support roadmap for 2025-2026. The target market for SwingVision includes over 170 million participants in amateur tennis, badminton, and racquetball in the U.S., but the company's primary focus has been on tennis.

## Racket Sports Expansion: Padel and Pickleball

Expansion into other racket sports, including padel, is on the company's roadmap. More specifically, "Coming to Padel this Fall" according to the YouTube channel information.

SwingVision intends to broaden its offerings to cover pickleball, one of America's fastest-growing sports. The new funding will unlock key hires to build tournament-grade officiating and remote coaching capabilities into the platform, and to expand beyond tennis into pickleball and padel.

The company's vision is broader than any single sport: The company's ultimate goal is to create a multi-sport platform that redefines how athletes of all levels record, share, and enhance their game experiences.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

18. user (2026-06-10T05:43:56.861Z)

## F<sup>3</sup>Set Paper Overview

**Title:** "F<sup>3</sup>Set: Towards Analyzing Fast, Frequent, and Fine-grained Events from Videos"

**Abstract Summary:**

The paper introduces F<sup>3</sup>Set, a benchmark dataset designed to address challenges in detecting events that are simultaneously fast, frequent, and fine-grained in video content. According to the abstract, the benchmark "consists of video datasets for precise F<sup>3</sup> event detection" and features "over 1,000 event types with precise timestamps."

The researchers evaluated existing temporal action understanding methods and found them lacking. They also propose F<sup>3</sup>ED, a novel detection approach that demonstrates improved performance.

**Regarding Racket Sports Specificity:**

The abstract notes that "F<sup>3</sup>Set contains several sports datasets" but does not explicitly limit the scope to racket sports specifically. The framework is described as potentially extensible to

other applications, suggesting broader applicability beyond any single sport category.

The work was accepted to ICLR 2025, and associated resources including datasets, code, and models are made publicly available through their GitHub repository.

## 19. user (2026-06-10T05:44:01.813Z)

Web search results for query: "India badminton AI video analysis app startup highlights rally 2025 -Machaxi -Stupa"

Links: [{"title": "Custom AI Badminton Video Analysis Software Development", "url": "https://tezeract.ai/industries/badminton-video-analysis-software/"}, {"title": "This Startup aims to assist badminton players with an AI coach", "url": "https://indiaai.gov.in/article/this-startup-aims-to-assist-badminton-players-with-an-ai-coach"}, {"title": "Play Badminton Like a Pro: AI-Enhanced Sports Learning Videos | ReelMind", "url": "https://reelmind.ai/blog/play-badminton-like-a-pro-ai-enhanced-sports-learning-videos"}, {"title": "10 Ways AI Is Being Used in Badminton [+5 Case Studies][2026] - DigitalDefynd Education", "url": "https://digitaldefynd.com/IQ/ai-use-in-badminton/"}, {"title": "IIIT-H Is Now Using AI For Analysing Badminton Games | AIM", "url": "https://analyticsindiamag.com/iiit-h-is-now-using-ai-for-analysing-badminton-games"}, {"title": "A Comparative Analysis of Deep Learning Models and Gradient Computation for Rally Detection in Badminton Videos | SN Computer Science | Springer Nature Link", "url": "https://link.springer.com/article/10.1007/s42979-025-03935-0"}, {"title": "Indian badminton players use artificial intelligence to up their game - Mugglehead Investment Magazine", "url": "https://mugglehead.com/indian-badminton-players-use-artificial-intelligence-to-up-their-game/"}, {"title": "Instant review system for badminton: computer vision system use case - Spyrosoft", "url": "https://spyro-soft.com/blog/artificial-intelligence-machine-learning/instant-review-system-for-badminton-computer-vision-use-case"}]

Based on the search results, here are some key findings about AI video analysis apps and badminton startups in India:

## AI Badminton Analysis in India

Birdies is a Badminton AI coach that helps players to understand the areas they are performing well and areas they have to focus more on. BirdieIns is available both in the B2B and B2C formats.

In the B2B space, BirdieIns provides comprehensive data along with auto-generated feedback, and has a provision for the coach to annotate video frames or any graph before sharing feedback data in PDF format. The company achieved more than 95% in service detection, 94% in shot detection, and 98% in player detection/speed.

## Adoption in India

Nearly 70 schools in Bengaluru alone have been getting an AI model to critique their form. This indicates significant adoption of AI analysis tools for badminton training in India.

## Academic Research

International Institute of Information Technology, Hyderabad (IIIT-H) successfully developed and tested a viable market prototype for badminton analytics, developed at its Center for Visual Information Technology in collaboration with Star Sports which saw the data going live during the telecast of the Premier Badminton League in January 2019.

The search results show robust development of AI-powered badminton analysis tools in India, though I did not find specific information about a startup highlights rally specifically in 2025.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

**20. assistant (2026-06-10T05:44:02.370Z)**

You've hit your session limit · resets 3:50pm (Asia/Calcutta)